Gömb tuning vs. non-HSiKAN AUC bars (SGCN territory)

ops

2026-05-12

Scope. Position HymeKo-Gömb (single-k, mixed-arity, and joint-mix joint-BA recipe) against *external* signed link-prediction bars already documented in-repo: **published SGCN** (Derr 2018, ~0.91–0.93 AUC on Bitcoin SNAP graphs) and **in-repo tuned SGCN** (e.g. Alpha 0.927, OTC 0.957, Slashdot ~0.914 under matched protocol; see `PAPER_DRAFT_2026_05_03.md`). HSiKAN headline numbers are *out of scope* as the primary target; they serve only as context.

Goal (staged). Stage A (publishable): maximize single-seed `test_auroc` from `run_gomb_tune` / `run_gomb_smoke` until best trial meets or exceeds published SGCN (≥0.91 Bitcoin Alpha, ≥0.93 Bitcoin OTC under our split). **Stage B (stretch):** reduce gap to in-repo tuned SGCN where VRAM and enumeration caps allow.

Affected files. `docs/plans/2026-05-12-gomb-external-auc-tuning/*`, `signedkan_wip/experiments/run_gomb_extern` `reports/2026-05-12-gomb-external-auc-tuning-results.md` (append-only JSONL summaries). **CORE.YAML:** none.

Procedure. (1) Joint-mix envelope: `run_gomb_tune --joint-mix` on `bitcoin_otc`, `bitcoin_alpha` (Bitcoin-first; VRAM clamps already in `for_joint_mix_tuning`). (2) Vanilla/mixed envelope: same driver *without* `--joint-mix` to explore single-k default Gömb and occasional `--cycle-ks 3,4`. (3) Log JSONL; compare best `test_auroc` to Stage A/B thresholds. (4) Optional later: Slashdot under `--architecture compact`.

Test strategy. No new unit tests required for the shell driver; existing `test_hymeko_gomb.py` tuner tests remain the regression gate for `run_gomb_tune`.

Performance budget. Per subprocess: existing `--timeout-s` default (7200 s); total wall for a dual-envelope batch ≤2 h on RTX-class 8 GB unless datasets expand.

Rollback. Remove plan directory and experiment script; delete or archive JSONL artifacts; no framework defaults change.

Risk. CUDA OOM on joint-mix despite clamps; single-seed maxima overfit val; Slashdot requires compact search and long epochs—defer to follow-up.

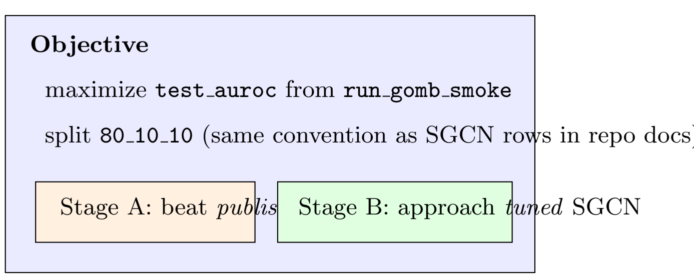

Figure 1: Objective: Gömb smoke output vs. external SGCN bars.